

Project Scope Overview

Develop and finalize backend functionalities in an existing Node.js project. This includes implementing defined routes, basic CRUD operations, ensuring security, and delivering complete and clear documentation for the entire backend system.

Core Functional Requirements

1. CRUD Operations

Implement Create, Read, Update, and Delete functionalities across specified modules:

- Products
- Links
- Learning modules

These are tied to a simple backend admin view (only accessible to system admins).

2. Messaging Module (Bulletin Board)

- Implement a non-real-time messaging feature (bulletin-style).
- Mostly static content addition, minimal delete/update logic.

3. Image Upload Support

- Backend should handle image uploads related to products and modules.

4. Schema-Adherence

- Critical Requirement: Use the predefined route schemas and request/response structure provided by the client.
 - Do not deviate from these schemas, as they are being used by the frontend team in parallel.
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Security Requirements

- Implement basic security measures such as:
 - Token access (already set up in the existing user module).
 - Rate limiting
 - General backend security hygiene
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Documentation Requirements

- Provide extensive documentation as you build each milestone:
 - API route documentation (with example requests/responses)
 - Clear structure of request/response bodies
 - Explanations of logic where relevant

Milestone-Based Development

- Work will proceed milestone-by-milestone.
- Proposal should be made for the first milestone initially.

Team Expectation

- Client prefers the same team to handle both backend and prediction model work to maintain consistency and avoid dilution from multiple teams.

Proposal Timeline

- Submit a detailed proposal by the next day after receiving the schema files and documents.

Overview

This milestone implements the Products, Services, and small LMS features outlined below.

Features

Products

1. API to return a list of system-defined products.
2. Backend GUI for super users to add/modify products.

Services

1. API with integration with Google Places API to search for businesses to return.
2. Should cache and rate limit results to reduce API costs.

LMS (Learning Management System)

1. API that returns Learning Modules to the client.

2. Backend GUI for super users to add/modify Learning Modules. Will need to support uploading images for the modules.
3. Learning Modules will just be HTML/Markdown content with images.